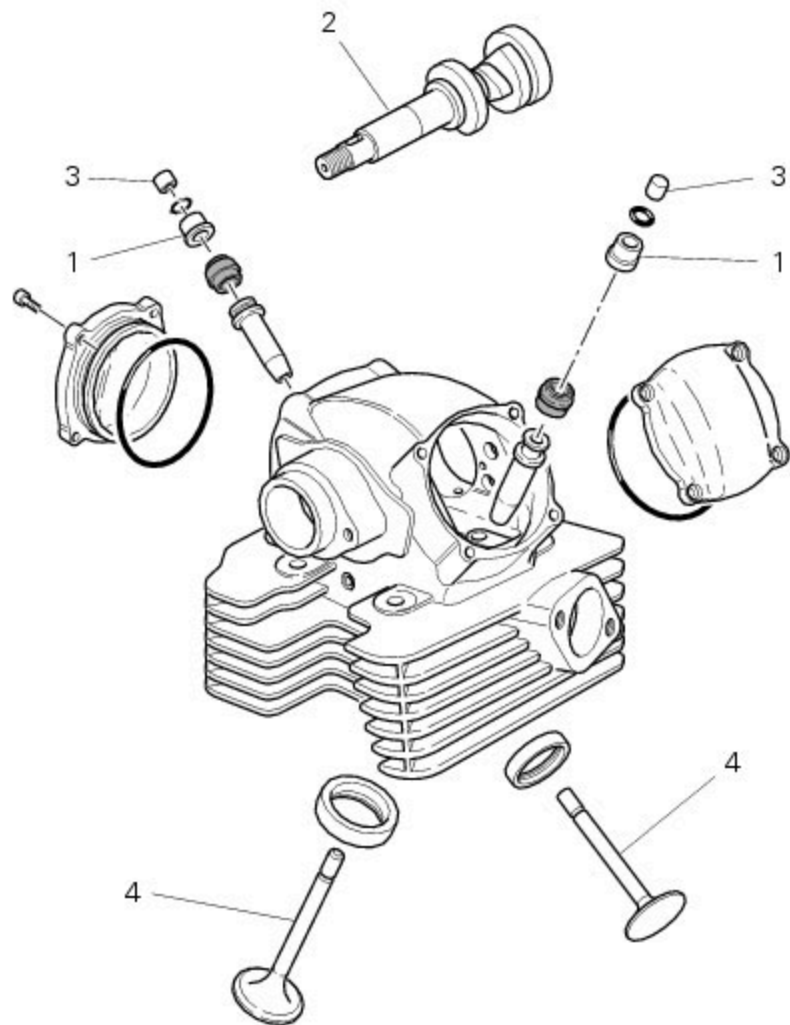


4.1 -Cylinder head assemblies: checks and adjustments



- 1 Closing shim
- 2 Camshaft
- 3 Opening shim
- 4 Valve

 Spare parts catalogue

1100 ABS	CYLINDER HEAD: TIMING SYSTEM
1100 ABS	VERTICAL CYLINDER HEAD
1100 ABS	HORIZONTAL CYLINDER HEAD
1100 S ABS	CYLINDER HEAD: TIMING SYSTEM
1100 S ABS	VERTICAL CYLINDER HEAD
1100 S ABS	HORIZONTAL CYLINDER HEAD

 Important

Bold reference numbers in this section identify parts not shown in the figures alongside the text, but which can be found in the exploded view diagram.

Checking and adjusting valve clearances

Operation	Section reference
Remove the seat	E 3, Removal of the seat

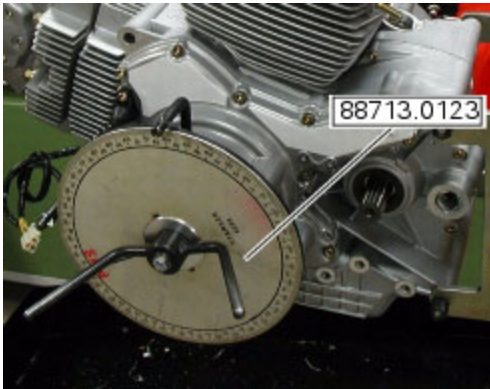
Remove the tank covers	E 2, Removal of the fuel tank fairings
Remove the fuel tank	L 2, Removal of the fuel tank
Remove the oil cooler from the frame	N 2.2, Removal of the oil cooler
Remove the valve covers	N 4.3, Removal of the valve covers



Note

For clarity, the figures show the engine block removed from the frame.

Take off the inspection cover on the alternator side and install service tool no. **88713.0123**.



Turn crankshaft with tool no. **88713.0123**, and set it so that the horizontal piston is at the TDC in the combustion stage, i.e. the valves of the horizontal head are all at rest position: check the valve clearance on the horizontal cylinder head. Reset the degree wheel of tool no. **88713.0123**.

To do the same on the vertical cylinder, turn crankshaft through 270° anticlockwise from current position, i.e. where tool no. **88713.0123** is reset. In this way the vertical piston is at the TDC in the combustion stage, i.e. the valves of the vertical head are all at rest position, and you can now check valve clearance on the vertical head.

Checking the opening clearance (Sa) of the intake/exhaust valves

To check the valve opening clearance (Sa), insert the blade of a feeler gauge between the opening rocker (A) and the shim (3).



Note

While inserting the blade of the feeler gauge to check the clearance, apply gentle leverage to overcome the force of the closing rocker spring and thus eliminate any play in the rocker mechanism: then check that the shim can rotate freely.

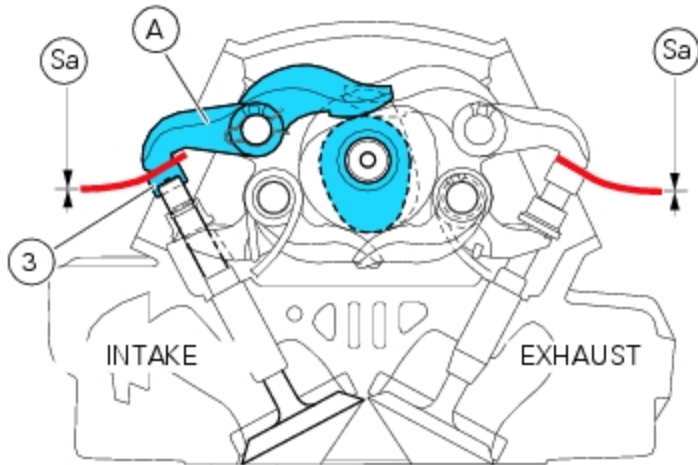
The clearance values must be within the specified limits (Sect. C 1.1, [Timing system/valves](#)).

If not, remove the opening shim (3), as described in chapter "[Removal of the rocker arms, shims and valves](#)", Sect. N 4.4, and replace it with one of suitable height to obtain the prescribed assembly clearance.



Note

Opening rocker arm shims measuring 1.8 to 3.45 are available as replacement parts: the size is punched on the shim.



Checking the closing clearance (Sc) of the intake/exhaust valves

To check the valve closing clearance, insert the blade of a feeler gauge between the closing rocker (B) and the shim (3): the closing clearance (Sc) corresponds to the thickness of the feeler gauge and should allow the free rotation of the shim without any significant play.

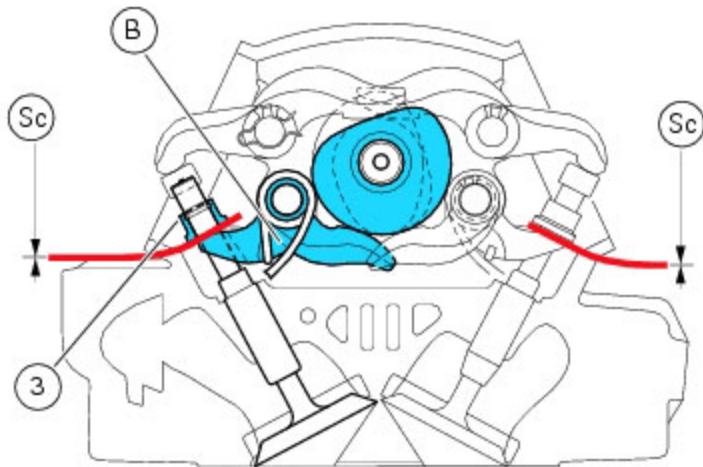
The value must be within the prescribed ones (Sect. C 1.1, [Timing system/valves](#)).

If not, remove the closing shim (1), as described in chapter "[Removal of the rocker arms, shims and valves](#)", Sect. N 4.4, and replace it with one of suitable height to obtain the prescribed assembly clearance.



Note

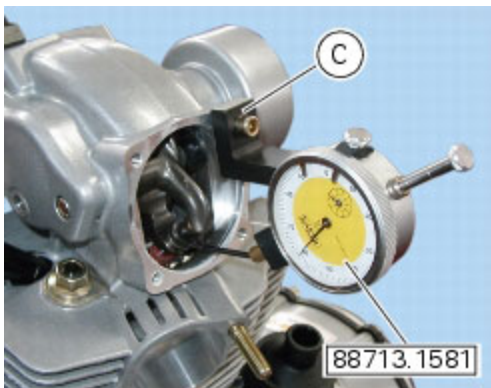
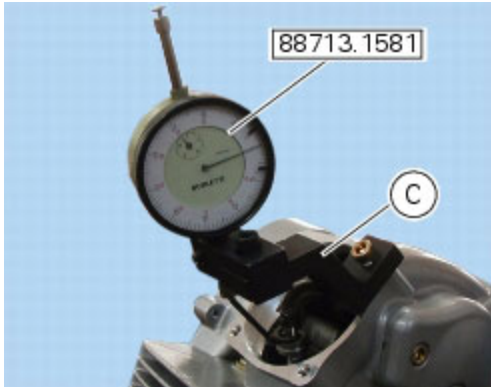
Closing rocker arm shims measuring from 2.2 to 4.5 are available as replacement parts: the size is punched on the shim.



Checking valve lift

Fit the gauge (C) **88765.1581** in correspondence with the fixing hole of the previously removed cylinder head valve cover, as shown in the photo.

Reset the valve opening clearance when the camshaft is in its rest position by inserting a feeler gauge between the upper rocker arm and the opening shim.



Note

This operation is best done using one dial gauge for the exhaust valve and another one for the intake valve.

Make sure the dial gauge fork is centred relative to the valve axis and located against the closing shim.

Set the dial gauge to zero when the valve is fully closed.

Rotate the intake camshaft so as to allow the intake valves to lift fully.

Check on the dial gauge that the measured value corresponds to the prescribed one (Sect. C 1.1, [Timing system/valves](#)).

Perform the same procedure for the exhaust valve, fixing the support to the opposite side of the cylinder head.

Remove the tools **88765.1581**.

Operation	Section reference
Refit the valve covers	N 4.3, Refitting the valve covers
Refit the oil cooler to the frame	N 2.2, Refitting the oil cooler
Refit the fuel tank	L 2, Refitting the fuel tank
Refit the tank covers	E 2, Refitting the fuel tank fairings
Refit the seat	E 3, Refitting the seat

Checking the engine timing

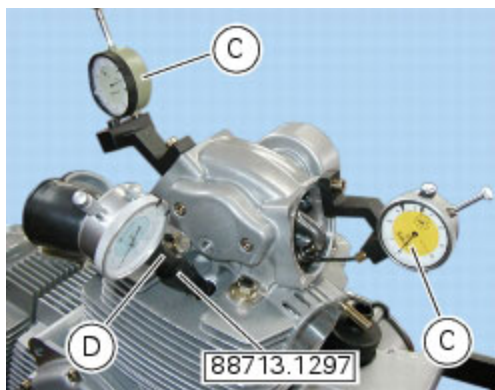
Operation	Section reference
Remove the seat	E 3, Removal of the seat
Remove the tank covers	E 2, Removal of the fuel tank fairings
Remove the fuel tank	L 2, Removal of the fuel tank
Remove the oil cooler from the frame	N 2.2, Removal of the oil cooler
Remove the valve covers	N 4.3, Removal of the valve covers
Remove the horizontal cylinder exhaust pipe	L 8, Removal of the exhaust system
Disconnect the timing-side spark plug cables	M 3, Spark plug
Remove the side timing belt covers	N 4.2, Removal of the timing belt covers

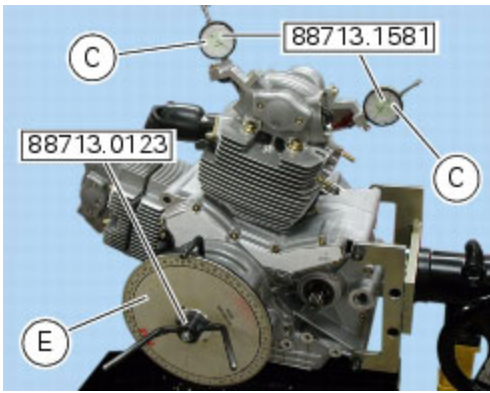


Note

Engine timing should be checked after changing parts such as camshafts, valves or timing belt rollers or when overhauling badly worn engines.

Remove the spark plugs and install the tool (D) **88765.1297** in the spark plug bore to determine the piston TDC, along with gauges (C) **88713.1581** and the timing check tool with timing degree wheel (E) **88713.0123**.





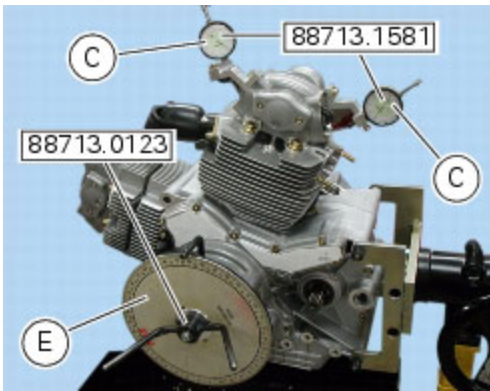
Reset the opening valve clearance when the camshaft is in rest position by inserting a feeler gauge between the opening rocker arm and the opening shim.

Check that in this condition the camshaft can rotate. If it moves stiffly, use a thinner feeler gauge.

In this condition, with the piston of the horizontal cylinder at TDC and the valves fully closed as confirmed by the reading on dial gauge (E), set gauges (C) to zero.

Set the tension value of the belts as described at Sect. D 5, [Checking and adjusting timing belt tension](#).

Turn the degree wheel (E) anticlockwise until the gauge dial (C), on the exhaust side, shows a lift of **1 mm**. Check that the value of the angular displacement read on the degree wheel (E) is as specified in (Sect. C 1.1, [Timing system/valves](#)).



Continue to rotate in the same direction until you obtain a lift of **1 mm** on the intake side. Check the angular value on the degree wheel.

Continue to rotate until, in correspondence with the closure of the valve for the compression stroke, you obtain a lift of the intake valve of **1 mm** on the gauge (C). Check the angular value with the prescribed one (Sect. C 1.1, [Timing system/valves](#)).

Rotate clockwise again until obtaining 1 mm lift of the exhaust valve, on both valve opening and closure.

Check the angular displacement value against the specified value.

Repeat the procedure for the vertical cylinder.

Permissible tolerance for measured values is $\pm 3^\circ$ with respect to specified values.

Remove the tools used to check engine timing.

Operation	Section reference
Refit the timing belt covers	N 4.2, Refitting the timing belt covers
Connect the timing-side spark plug cables	M 3, Spark plug
Refit the horizontal cylinder exhaust pipe	L 8, Refitting the exhaust system
Refit the valve covers	N 4.3, Refitting the valve covers
Refit the oil cooler to the frame	N 2.2, Refitting the oil cooler
Refit the fuel tank	L 2, Refitting the fuel tank
Refit the tank covers	E 2, Refitting the fuel tank fairings
Refit the seat	E 3, Refitting the seat

